Bogleheads

Using a spreadsheet to maintain a portfolio

*This article contains details **specific to United States (US) investors**. Parts of it do not apply to non-US investors.*

You should look at all of your accounts as a unified portfolio to construct a portfolio that is low cost, well diversified, and tax efficient. If this concept is new to you, you might feel overwhelmed by the complexity of maintaining such a portfolio. **Using a spreadsheet to maintain a portfolio** is intended to suggest one way of making things easier.

Asset allocation

One of the easiest ways to maintain your portfolio is to categorize by asset allocation. For example:

Account	Domestic stocks	REITs	International stocks	Bonds	Total
Taxable	-	-	$14,000	-	$14,000
Roth IRA	-	$9,000	$9,000	-	$18,000
401(k)	$49,000	-	-	$19,000	$68,000
Total	$49,000	$9,000	$23,000	$19,000	$100,000
Desired	$48,000	$8,000	$24,000	$20,000	$100,000
Difference	($1,000)	($1,000)	$1,000	$1,000	-

- You manually enter cells in the rows for Taxable, Roth IRA, and 401(k). All other cells can be automatically calculated.
- Each cell in the Total row represents the total amount of domestic stocks, that of real estate investment trusts (REITs), and so on.
- The Desired row represents the desired amount of dollars in each asset class. You can calculate this by multiplying the value of your portfolio by the percentages in your target asset allocation.
- The Difference row tells you how far off your portfolio is relative to your desired asset allocation. You can calculate it by subtracting a cell in the Total row from the corresponding cell in the Desired row.

Sample spreadsheets

- Here is a sample spreadsheet (https://docs.google.com/spreadsheets/d/1E28C1YTb1jzi3hSKK3ABGvWgTmjSlO5W/edit?usp=sharing&ouid=103809420707053067543&rtpof=true&sd=true) and documentation (https://drive.google.com/file/d/1AdNNA8LuYbRyeomrjqLrj9rOvnlWF_xE/view?usp=sharing) in Google Drive in this style. It includes the additional features above, and can also adjust for the different after-tax value of assets in different accounts, and combine multiple subclasses into one class (so that you may have a target allocation to foreign stocks, and separate targets for developed and emerging markets). A simpler version (https://docs.google.com/spreadsheets/d/1jY8AdbIF2Xm4Tdq_z2YAf2Zxd42xxUvh/edit?usp=sharing&ouid=103809420707053067543&rtpof=true&sd=true) is also available. This version does not track subclasses.

- One important aspect of maintaining your asset allocation is to do periodic rebalancing. A sample rebalancing spreadsheet is available from Google Drive (https://docs.google.com/spreadsheet/ccc?key=0AsVSNcUR-B6IdE4tZm0yVk9ObkxsZ2xvNmJpRTB0b0E) (by forum member LadyGeek).[note 1] It uses three example approaches, which you could incorporate into any of the sample spreadsheets.
- Asset Allocation Spreadsheet (https://docs.google.com/spreadsheet/ccc?key=0AueLhNpKZa4XdG4wUEwwU3BmQlNqVkc0cVpLNmhqX1E) - A Google spreadsheet created by Hoppy08520, with a link to documentation, that can help setup a multi-account portfolio.
- Portfolio Tracker (https://github.com/danbuchal/portfolio-tracker/releases) - A Google spreadsheet produced by forum poster 'TrackerDan', with several ways of visualizing your allocation, and near-real-time value updates. Support is in Bogleheads forum topic: "A Free Google Sheet for Asset Allocation Tracking" (https://www.bogleheads.org/forum/viewtopic.php?t=451043).

Suggested improvements

- You might want to make a separate Target row to keep track of your target allocation. In that case, the Desired row would be the product of the Target row and the total portfolio.
- Similarly, you might make an Actual row to compute your percentage allocation, so that you could compare your target percentages to your actual percentages.
- If you have funds which cover multiple asset classes, you can put each fund in a different row, make an extra column for the total fund value, and have the spreadsheet automatically compute the amount of the fund which is in each class. This lets you easily track the effect on your allocation of adding $10,000 to a balanced fund.

Asset allocation visualized

The Finance Buff's article, Cascading Asset Allocation Method (http://thefinancebuff.com/cascading-asset-allocation-method.html), describes how to select a portfolio's asset allocation by visualizing the process as a top-down hierarchy. The most important selection is at the top (stocks/bonds), which then breaks down into smaller asset classes. Selection of the actual funds is done last, which is at the bottom of the chart. Two spreadsheets are available.

First, forum member 'pradador' has created a spreadsheet which lets you create a portfolio by following the advice in the article. The spreadsheet is available on Google Drive: Cascading Asset Allocation (https://docs.google.com/spreadsheet/ccc?key=0AsVSNcUR-B6IdEJZNE43NHpKQnNPNXFUTk1CZ195dmc) (The spreadsheet was further modified by forum member LadyGeek to include error checking.)[1]

This spreadsheet includes a fixed income major asset category, which is not mentioned in the article. There is no consensus on the role of fixed income in a portfolio; this depends on your situation. If you consider *fixed income* as a bond, then set the percentage of Fixed Income to 0.

There are two tabs:

- Top Down: Starting on the left side and proceed to the right, enter your desired asset allocations to arrive at the percentage of each fund's contribution to your portfolio. The bottom row is reserved for error checking. If you make an entry (the percentages do not add up to 100%), you will see an error message.

- Bottom Up: Use this tab to determine the asset allocation percentages of your portfolio when you know the funds. In the right-most column, enter the percentages and funds. The spreadsheet will work left and arrive at the asset allocations. The blank cell in the lower right-most corner will display an error if the percentage does not add up to 100%.

How to use a proxy fund

Google Finance offers historical and real-time securities information using the `GoogleFinance()` function in Google Sheets.

If you have funds held in a Collective Investment Trust (CIT), in which there is no ticker symbol, then you can use a proxy fund to keep your balance roughly up to date using the GoogleFinance functions in Google spreadsheets.[note 2]

For example, suppose you hold the government's Thrift Savings Plan C Fund, which is a collective investment trust (CIT) that tracks the S&P 500 index. Find out the closing price of the CIT on the current day, and also find a similar proxy fund, such as Vanguard's institutional index fund (VINIX).

For example, suppose the C Fund's closing price is $23.6602 and Vanguard's VINIX is $167.75. This means that a share of the C Fund is worth 7.08 times a share of VINIX (167.75/23.6602).

To maintain a real-time estimate of your C Fund balance without needing to update the C Fund's share price every day, you can instead take the daily price of VINIX and divide it by the multiplier to get an approximate share price for the C Fund.

For example, these could be the cell along a row in your spreadsheet and this is what you would enter into them.

Implementing a proxy fund

Cell	Heading	Enter	Notes
A2	Shares	100	Enter the number of shares you have in the Collective Index Trust (CIT). As you add or subtract shares from contributions, withdrawals or exchanges, you will need to periodically update this amount.
B2	Closing Price	23.6602	Enter the CIT's closing price. You will need to look this up in your plan.
C2	Close Date	01/03/2014	Enter the closing date on which you got the "Closing Price" of the CIT in B2. Make sure you enter the date and closing price are in sync, and be sure that the closing price has been updated in case there is a lag between the market close and the posting of the closing prices.
D2	Proxy Fund	VINIX	Enter the ticker of a fund that is similar to your CIT.
E2	Proxy Price	=INDEX(GoogleFinance(D2, "close", C2); 2; 2)	This returns the closing price of the proxy fund you selected in D2, on the date C2, which is the same date you recorded the closing price of the CIT.
F2	Multiplier	=E2/B2	This is the ratio of the Proxy Fund's Price divided by your CIT's price, on the same date.
G2	Proxy Fund Current Price	=GoogleFinance(D2, "price")	The current price of your proxy fund.
H2	CIT Approx Price	=G2/F2	This is the approximate calculated share price of the CIT.
I2	Balance	=H2*A2	The approximate balance you hold in the CIT. This is the approximated price of the CIT multiplied by the number of shares you hold in the CIT.

To keep this reasonably up-to-date and to prevent the "multiplier" from drifting over time, you can periodically replace the date in cell C2 and the actual closing price of the CIT in cell B2.

Real-time tracking from a website

Another technique is to find a website that outputs the price of the trust in real-time (or near real-time) and then use the GoogleFinance `ImportHmtl()` function to "scrape" the HTML from the page.

If the `ImportHtml()` function can scrape that data cleanly into defined rows and columns with the price landing in a single cell, then you can then use the `INDEX` function to get the value of a particular cell by referencing its row and column. The Bogleheads forum shows two examples:

- Bogleheads forum topic: "GoogleFinance price quotes for TSP Funds" (https://www.bogleheads.org/forum/viewtopic.php?t=141547)
- Bogleheads forum topic: "How to import TIAA-CREF annuity unit values into Google Docs" (https://www.bogleheads.org/forum/viewtopic.php?t=141501)

Rate of return

See also: Comparing investments § Internal rate of return, and Calculating personal returns

Calculating personal returns offers a spreadsheet to calculate investor and portfolio returns. It lets Bogleheads report their personal returns in a uniform manner.

If you know the dates you made your investment contributions and your withdrawals, the XIRR (Internal Rate of Return) spreadsheet function is the best and most accurate method for calculating your rate of return. This method is far more accurate than annual average returns, as contributions and withdrawals (cash flows) are weighted for both time and value.[2][note 3]

Bogleheads forum topic: "XIRR function again" (https://www.bogleheads.org/forum/viewtopic.php?t=115030) has details and a working example.[note 4]

Tax lot tracking

Another use of a spreadsheet is to keep track of share lots in a taxable account, so that you can automatically compute the capital gain or loss when you sell, and easily see which shares should be sold for tax loss harvesting or to minimize the tax cost of selling. A spreadsheet for tax lots could look like this:

Date bought	Amount invested	Share price	Shares	Date sold	Share price	Amount received	Capital gain
2/1/08	$5000	$20.00	250.000	10/31/08	$15.00	$3750	-$1250
5/1/08	$5000	$18.00	277.778	-	-	-	-
7/1/10	$10000	$30.00	333.333	2/1/11	$33.00	$11000	$1000

Sample spreadsheets

- Here are three versions of a spreadsheet for tracking tax lots, and computing the capital gains and losses on each sale. There is a version for stocks or ETFs (https://web.archive.org/web/20190117214306/http://remarque.org/~grabiner/stockworksheet.xls), most mutual funds (https://web.archive.org/web/20190225201418/http://remarque.org/~grabiner/mutualfundworksheet.xls), and funds with purchase and redemption fees (http://remarque.org/~grabiner/fundwithfees.xls). In a brokerage account, you would make one sheet in a workbook for each holding. The worksheets adjust for stock splits and share class conversions, but if you sell part of a lot, you have to split the lot into two purchases manually.

Portfolio tools

Main article: Portfolio tools

In addition to the spreadsheets there are various utilities to design and manage portfolios that span multiple accounts. Most tools cannot automatically compose your ideal portfolio, but you can use the tools to enter two or three models that you composed, and then you can compare and contrast the results, looking in particular at the blended expense ratios and the totals of each asset class.

Vanguard Portfolio Watch is a popular online tool for Vanguard clients to track asset allocation.

Morningstar's *Instant X-Ray* portfolio analysis tool has been discontinued effective April 17, 2025.[3] Forum member LadyGeek has developed a spreadsheet alternative. See Bogleheads forum topic: "Wiki - Morningstar's free Instant X-ray tool no longer available - Alternatives?" (https://www.bogleheads.org/forum/viewtopic.php?t=469099)

Notes

1. To download, select File --> Download As --> Excel or OpenDocument.
2. Proxy fund: A fund used to represent another fund. The C-fund does not have a ticker symbol. In its place, the Vanguard Institutional Index Fund (VINIX) is used to represent the C-fund. The Vanguard fund is the proxy for the C-fund.
3. *Internal* refers to the fact that its calculation does not incorporate environmental factors, for example, the interest rate or inflation. For more, see Internal rate of return, from Wikipedia
4. Bogleheads forum topic: "XIRR function again" (https://www.bogleheads.org/forum/viewtopic.php?t=115030) also describes a bug in MS Excel's `XIRR()` function. This bug is not present in LibreOffice or Google Drive Sheets.

See also

- Tools and calculators
- Rebalancing

References

1. Bogleheads forum topic: "Cascading Asset Allocation Spreadsheet" (https://www.bogleheads.org/forum/viewtopic.php?t=106505), pradador. December 5, 2012.
2. Bogleheads forum topic: "Using XIRR in Excel" (https://www.bogleheads.org/forum/viewtopic.php?t=51221). March 3, 2010.
3. "Upcoming changes to our membership offerings, tools, and features" (https://global.morningstar.com/en-gb/updates/upcoming-changes-our-membership-offerings-tools-features). *Morningstar*. April 15, 2025. Retrieved March 10, 2026.

External links

- LibreOffice Calc (https://www.libreoffice.org/discover/calc/), a free open-source spreadsheet that's compatible with Microsoft Excel. Available for Microsoft Windows, Linux, and Mac OS X.

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